

ARM0: A Six-Degree-of-Freedom Robotic Arm

Embodied Intelligence Group

University of Florida

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Abstract

The Embodied Intelligence Group (EIG) is a student organization at the University of Florida focused on robot learning. Through ARM0, we aim to develop a six-degree-of-freedom (6-DOF) robotic arm that will serve as a hardware platform for future robot learning projects. Six-axis arms are a common industry standard for general-purpose manipulation: six independent degrees of freedom are the mathematical minimum needed to control a tool's full position and orientation in three-dimensional space [1]. This enables a suitably designed arm to achieve arbitrary poses within its reachable workspace, subject to joint limits and singularities. ARM0 will give members experience developing an integrated robotic system while establishing a foundation for learning-based manipulation.

1 Objectives and Deadlines

Deadline	Milestone and deliverables
December 11, 2026	Design V1 complete. Complete CAD assembly with all parts and subsystems designed, electronics configured, and software tested on another robotic arm.
April 30, 2027	Project 100% complete. Assembled and tested arm, demonstration against agreed requirements, and build, operating, and maintenance documentation.

Semester deadlines align with the end of fall 2026 and spring 2027 final exams in the [UF academic calendar](#).

2 Teams and Areas of Work

ARM0 will bring together three teams responsible for the arm's structure, electronics, and control software. Members will help shape the design and coordinate across teams as component choices and requirements develop.

- **Mechanical design.** Develop the arm's physical structure, including links, joints, and component mounts. Work includes CAD modeling, mechanism and material selection, load analysis, prototyping, and assembly. The team considers how reach, stiffness, weight, and manufacturability affect the arm's performance.
- **Electrical design.** Develop the power and signal systems that connect the controller, motors, and sensors. Work includes component selection, circuit schematics, wiring, motor-driver integration, and PCB design where needed. The team handles electrical testing and troubleshooting, with attention to power requirements, protection, and reliable communication.
- **Software design.** Develop the software that commands the arm and coordinates its motion. Work includes embedded programming, sensor feedback, kinematics, motion control, and simulation. The team also builds interfaces for operating the arm and recording data, supporting integration with future robot learning projects.

3 Related Robotic Arms

These platforms inform our research into construction, controls, and documentation.

- **LeRobot SO-101:** Five arm joints plus a gripper, with a leader–follower setup and assembly guides. It illustrates accessible hardware and teleoperation [2].
- **Seeed Studio reBot B601:** Six arm DOF plus a gripper, with published hardware and software that support reproducible development [3].
- **Annin Robotics AR4:** A six-axis arm with build files and control software, offering insight into assembling and supporting a member-built system [4].
- **UFACTORY xArm6:** A commercial six-axis system that provides an example of integrated operation and usability [5].
- **ROBOTIS OpenMANIPULATOR-X:** Four arm DOF plus a gripper, providing a simpler comparison of joint count and task capability [6].

4 Useful Skills

Beginners are welcome. Experience in the following areas is useful; named languages and tools are examples rather than prerequisites.

- **Mechanical design:** CAD modeling and assemblies in SolidWorks, Onshape, or Fusion; technical drawings, basic load analysis, and fabrication.
- **Embedded programming:** C/C++, microcontroller programming, interfacing with motors and sensors, and debugging hardware–software communication.
- **Electronics:** Circuit design, schematics and PCB layout in tools such as KiCad; wiring, soldering, power systems, and electrical troubleshooting.
- **Robotics and controls:** Python or MATLAB, coordinate transforms, kinematics, feedback control, and simulation; familiarity with ROS is also useful.
- **Testing and collaboration:** Git/GitHub, Linux, writing test scripts, analyzing data, and documenting designs and experiments.

5 Living Document

This is a living document: all content, including goals, deadlines, and team responsibilities, is subject to change. We will update it as the project progresses and our requirements, designs, and resources develop.

References

- [1] K. M. Lynch and F. C. Park. *Modern Robotics*. Chapter 2: Configuration Space.
- [2] Hugging Face. *LeRobot: SO-101*.
- [3] Seeed Studio. *reBot-DevArm: Hardware and Software*.
- [4] Annin Robotics. *AR4 Robot Arm*.
- [5] UFACTORY. *xArm5, xArm6, and xArm7 Comparison*.
- [6] ROBOTIS. *OpenMANIPULATOR-X*.